

# QI SHAO

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Harbin Engineering University, No. 145, Nantong Street, Nangang District, Harbin, Heilongjiang Province, China

## EDUCATION

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| <b>University of Liverpool</b> , Liverpool, United Kingdom                                                                                                                                                         | 08/2025 – 08/2027 [Expected] |
| • <i>Computer Science</i>   <i>Mentor: Prof. Baoru Huang, Ph.D.</i>                                                                                                                                                |                              |
| <b>Harbin Engineering University</b> , Heilongjiang, China                                                                                                                                                         | 09/2023 – 06/2028 [Expected] |
| • <i>PhD of Control Science and Engineering</i>   <i>Mentor: Prof. Liang Li, Ph.D.</i>                                                                                                                             |                              |
| <b>Peking University Summer School</b> , Beijing, China                                                                                                                                                            | 06/2020 – 08/2020            |
| • <i>Participated in courses related to Spatiotemporal Intelligence</i>   <b>GPA: 85/100</b>                                                                                                                       |                              |
| • <i>Relevant courses: GNSS-based Spatiotemporal Intelligence</i>                                                                                                                                                  |                              |
| <b>Harbin Engineering University</b> , Heilongjiang, China                                                                                                                                                         | 09/2019 – 06/2023            |
| • <i>Bachelor of Science: Measurement and control technology and instruments</i>   <b>GPA: 83.66/100 (13/122)</b>                                                                                                  |                              |
| • <i>Relevant courses: Automatic Control Theory, Robot Control, Instrument Measurement and Control, Artificial Intelligence (including reinforcement learning algorithms and digital image processing methods)</i> |                              |

## RESEARCH EXPERIENCE

### [Multi-source Data Fusion]

*Supervised by Professor Kunpeng He | Funded by China State Shipbuilding Corporation Ltd.*

#### **Project1: Research on Gravity and Terrain Matching Navigation**

- Motivation: To enhance positioning accuracy and reliability in GPS-denied or interfered environments (e.g., underwater, complex terrain) by designing an integrated navigation system based on geophysical field matching.
- Achieved a 12% improvement in positioning accuracy through the integration of these algorithms.

*Supervised by Professor Liang Li | Project in GNSS Precision Navigation Technology Lab*

#### **Project2: Research on Multi-Source Ultra-Rapid Satellite Clock Bias Product Combination Estimation Methods**

- Motivation: Address satellite clock bias discontinuities at boundaries and combine multi-AC products to enhance clock accuracy and consistency.
- Algorithm: Designed a terrain/marine gravity field-based matching positioning system; Proposed an improved Iterative Closest Contour Point (ICCP) algorithm and a Bayesian estimation-based gravity-terrain integrated navigation algorithm.
- Results: Experiments verified that combined gravity/terrain matching positioning outperforms single-source matching (gravity-only or terrain-only), improving navigation accuracy by 12% compared to traditional methods.

*Supervised by Professor Yonggang Zhang | Project in Intelligent Navigation and Information Fusion Lab*

#### **Project3: Factor Graph Optimization-Based Inertial-Vision-LiDAR Autonomous Localization and Perception System**

- Implemented a tightly coupled Inertial-Vision-LiDAR system and performed SLAM backend optimization using factor graph techniques.
- Designed and implemented a spatiotemporal alignment algorithm for calibrating and correcting time offsets between IMU and Visual Odometry (VO) data.
- Developed an NDT-based point cloud registration strategy for aligning heterogeneous data..

### [Computer Vision]

*Supervised by Professor Bing Li | Project in Smart Agricultural Machinery Technology Lab*

#### **Project4: Precise Visual Segmentation of Small Targets for Germ Rice Production**

- Motivation: Addressed the challenge of precisely identifying and segmenting small targets under complex conditions (adhesion, density) for germ rice production.
- Algorithm: Optimized Mask R-CNN by integrating Path Aggregation Network (PANet) and Efficient Channel Attention (ECA) to enhance feature extraction.
- Results: Achieved 98.5% segmentation accuracy, with mIoU >90% in key regions (e.g., germ), enabling pixel-level separation of complex adhered objects.
- enhanced feature extraction and recognition capabilities for small targets in complex scenes.

## PUBLICATIONS

**Q. Shao**, "Automatic Case Generation of Variation Testing in Navigation Software Based on the Genetic Algorithm," in Proc. 2023 Int. Conf. Mechatronics Autom. (ICMA), Aug. 2023, pp. 214-219. doi: 10.1109/ICMA57695.2023.10215844

## PATENTS

**Qi Shao**. "A multi-angle adjustable mounting bracket for measurement and control instruments." China Utility Model Patent CN 213207102 U. Issued May 14, 2021.

## TEACHING EXPERIENCE

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### *Teaching Assistant, Automatic Control Theory [Undergraduates Courses]*

02/2023 – 06/2023

- Guided students on principles, procedures, and data handling for core experiments, including: inverted pendulum (modeling, stability analysis, controller design) and servo systems (characterization, PID tuning, performance evaluation).

### *Teaching Assistant, Intelligent Vehicle Design and Implementation [Summer Courses]*

07/2024 – 08/2024

- Guided students through the core aspects of intelligent vehicles: perception, decision-making, and control. Assisted them in applying theoretical knowledge to practical systems, specifically through projects utilizing embedded systems, various sensors, and software platforms like ROS, Rviz, and Gazebo.

## HONORS

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- **First-Class Academic Scholarship** for Doctoral Students, Harbin Engineering University (Rank: 1/188) 2024-2025
- **Innovation Pacesetter Award**, Harbin Engineering University (Top 1% distinction; Nominated for the Chen Geng Scholarship, Harbin Engineering University's highest scholarship) 2022, 2024
- Team Leader, **National First Prize**, "Challenge Cup" National College Student Extracurricular Academic Science and Technology Contest (Set by China National Grain Group Co., Ltd.) 2024
- **National Gold Award**, China International College Students Innovation Competition 2023
- **National First Prize**, "Huawei Cup" China Postgraduate Artificial Intelligence Innovation Competition 2024
- **Finalist Award**, Mathematical Contest in Modeling (MCM) 2022
- Huawei Outstanding Student Developer, Huawei Developer Evangelist 2024

## RESEARCH INTERESTS

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### **[Medical Image Segmentation]**

- Investigating hybrid CNN-Transformer architectures like AD2Former for abdominal tumor segmentation, focusing on enhancing local and global feature integration.
- Exploring data-efficient Transformer models such as MedFormer for scalable segmentation across CT and MRI modalities.
- Collaborated with a tumor hospital on preoperative segmentation of abdominal tumors, aiming to provide guidance for clinical treatment planning.

### **[Robotics]**

- Developing real-time 3D scene reconstruction and dynamic environment understanding for robotic navigation.
- Designing human-robot collaboration frameworks for robots, integrating visual and tactile feedback for precise task execution.
- Investigating the application of AR/VR technologies to enhance visualization and decision-making.

## SKILLS

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- Programming: Python, C++, MATLAB
- Tools & Platforms: ROS (Rviz, Gazebo), PyTorch/TensorFlow (if applicable), OpenCV (if applicable), Git
- AI & Machine Learning: Deep Learning (CNNs, Transformers, Mask R-CNN), Genetic Algorithms, Bayesian Estimation
- Navigation & Fusion: Multi-Sensor Fusion (IMU, Camera, LiDAR, GNSS, etc.), Calibration, GNSS Data Processing, Inertial Navigation, Geophysical Field Matching
- Robotics & Control: Control Systems Theory & Design (PID), System Modeling, Sensor Integration, HRI
- Languages: Chinese (Native), English (Fluent/Proficient)